

CRM System

Software Requirements Specification

SRS v1.0

Document Status	Baselined for architecture. Supersedes working drafts v0.1–v0.3.
Prepared for	Ethiopian IT Park — CRM Project Stakeholders
Purpose	Baseline requirements reference for system architecture, database design, API design, and implementation
Distribution	Ethiopian IT Park CRM project stakeholders and engineering team

Document Status: Baselined for architecture. Supersedes working drafts v0.1–v0.3.

Legend (used throughout):

Tag	Meaning
[REQ]	Confirmed source requirement — traceable verbatim to Ethiopian IT Park's original CRM requirements document
[DEC]	Approved business decision — resolves an ambiguity or gap in a way that changes system behavior, approved by the business owner
[DEF]	Approved working default — reversible engineering/configuration choice, not a business-approved requirement
[ASM]	Assumption — adopted to proceed; not yet confirmed by an authoritative source; explicitly reversible
[EXT]	External dependency — requires input from a party outside this specification effort
[OOS]	Out of scope

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1. Introduction

1.1 Purpose

This document specifies the requirements for a Customer Relationship Management (CRM) system for Ethiopian IT Park. It is the baseline requirements reference for system architecture, database design, API design, and implementation. **[REQ]**

1.2 Intended Audience

System architects, database designers, backend and frontend engineers, QA engineers, and Ethiopian IT Park stakeholders responsible for approving system behavior.

1.3 Definitions and Terminology

Term	Definition
Lead	An unqualified prospective relationship captured from any source. [REQ]
Opportunity	A qualified, trackable engagement derived from a Lead, representing a commercial/business relationship, with stage, value, probability, and closing date. [REQ + DEC]
Account	An organization: startup, investor, company, government, or partner. [REQ]
Contact	An individual associated with an Account. [REQ]
Owner	The current user responsible for a Lead, Account, Opportunity, or Contact, for visibility purposes. [DEC] — this term is introduced by this specification; the source document only uses the verb "assign."
Assignee	The current user responsible for a Ticket or Activity — functionally the same mechanism as Owner, named separately to match the source's own language for these modules. [DEC]
Activity	A call, meeting, email, or task linked to a Lead, Account, Contact, or Opportunity. [REQ]
Ticket	A customer support/service request. [REQ]
Priority (Ticket)	Urgency level of a Ticket: Urgent, High, Medium, or Low. [REQ]
Role Tier	The relative seniority/oversight level of a CRM role (e.g., CRM Manager is the highest-tier role). Deliberately renamed from the source's use of "priority" to avoid collision with Ticket Priority. [DEC — terminology only, no behavior change]
Inactivity Duration	Elapsed time since a Ticket's last recorded activity. [REQ]
Reminder Threshold	Inactivity duration at which the Ticket's assignee is reminded. [REQ]

Term	Definition
Escalation Threshold	Inactivity duration at which a Ticket automatically escalates. [REQ]
Escalated	A flag on a Ticket indicating it has crossed its escalation threshold; independent of Ticket status. [DEC]
Program Application	A single entity representing a stakeholder's participation in a Program (incubation, training, workshop), carrying a status life-cycle (Applied → Accepted/Enrolled → Completed/Withdrawn). [REQ + DEC]
PMS	Tenant/Property Management System — the external system that manages office allocation, leases, and billing; out of scope for this CRM. [REQ]
Closed Won / Closed Lost	Terminal states of an Opportunity. [REQ]

1.4 References

- Source document: Ethiopian IT Park CRM Software Requirements Specification (original business-provided document; undated, unversioned).

2. System Overview

The CRM system provides Ethiopian IT Park with a centralized platform to manage relationships with startups, investors, partners, and corporate clients. It supports lead management, opportunity and pipeline tracking, account and contact management, activity and communication logging, support ticketing with inactivity-based escalation, and program/event participation tracking. **[REQ]**

The CRM is deliberately scoped apart from the existing Tenant/Property Management System (PMS): office allocation, lease management, and billing remain the PMS's responsibility. The CRM supports a manual, file-based data handoff to the PMS when an Opportunity is marked Closed Won. **[REQ]**

3. Stakeholders and User Roles

3.1 Stakeholders

Stakeholder	Interest
CRM Manager	Oversight, reporting, data quality [REQ]
Business Development Officer (BDO)	Daily lead/opportunity/account operations [REQ]
Support Agent	Ticket handling [REQ]
System Administrator	Configuration, security, integration settings [REQ]
Ethiopian IT Park management	Business outcomes and reporting visibility [DEC — inferred stakeholder, not a system user]

Stakeholder	Interest
PMS system owners	Recipients of the Closed-Won data handoff [REQ]

3.2 User Roles and Responsibilities

CRM Manager (highest Role Tier) — full visibility across all Leads, Opportunities, Accounts, Contacts, and Tickets; oversees pipeline and ticket-escalation performance; assigns tasks; accesses all reports and dashboards. **[REQ]**

Business Development Officer (BDO) — creates and manages Leads; converts Leads to Opportunities and Accounts; manages Accounts and Contacts; updates Opportunity stages; logs Activities and Communications. **[REQ]**

Support Agent — creates, views, and manages Tickets within their assignment and the shared support queue; updates Ticket status; communicates with customers; can manually escalate a Ticket. **[REQ]**

System Administrator — manages users and roles; configures pipeline stages, ticket categories, and escalation rules; manages reference/master data; manages export/integration configuration; maintains system security settings. **[REQ]**

3.3 Role-Based Visibility Model **[DEC]**

Role	Default record visibility
CRM Manager	Full visibility of all business records
Business Development Officer	Ownership-based — records for which they are the current Owner
Support Agent	Tickets assigned to them plus the shared support queue; read-only visibility of the Account and Contact linked to any Ticket they can see (does not cascade to that Account's Leads, Opportunities, or Activities)
System Administrator	Configuration and user-management access; not automatic business-record visibility. May configure export field mappings without needing visibility into actual exported data content.

Authorization is implemented as a resolvable policy layer (role grants + ownership, composable) rather than hardcoded per-query visibility rules, so that team-based or other elevated-visibility grants can be added later without redesigning the data model. **[DEC]**

Ownership-transfer control **[ASM, security-relevant]:** only a record's current Owner, a CRM Manager, or a System Administrator may reassign its ownership. A user cannot unilaterally assign an unowned or another user's record to themselves, to prevent using reassignment as a means of bypassing ownership-based visibility.

4. System Scope

4.1 In Scope

- Lead capture, qualification, and conversion **[REQ]**
- Opportunity and pipeline management, including the Closed-Won/Closed-Lost lifecycle **[REQ]**
- Account and Contact management **[REQ]**
- Activity and Communication tracking **[REQ]**
- Support ticketing with priority-based triage and inactivity-based escalation **[REQ]**
- Program and Event participation tracking **[REQ]**
- Reporting and dashboards **[REQ]**
- Role-based access control **[REQ]**
- Manual, file-based data export to the PMS on Closed-Won **[REQ]**

4.2 Out of Scope

See §14.

5. Functional Requirements

5.1 Lead Management

ID	Requirement
FR-LEAD-001	The system shall allow authorized users to create, view, update, and delete Leads. [REQ]
FR-LEAD-002	The system shall capture, at minimum, Lead name, organization, contact information, source, and category. [REQ]
FR-LEAD-003	The system shall allow assignment of a Lead to a staff member. [REQ]
FR-LEAD-004	The system shall track Lead status through at least: new, contacted, qualified, rejected. [REQ]
FR-LEAD-005	The system shall allow conversion of a qualified Lead into an Account, a Contact, and — where the engagement is commercial in nature — an Opportunity. [REQ, reading clarified by DEC]
FR-LEAD-006	The system shall warn (not block) on likely duplicate Leads based on matching email or phone. [DEF]
FR-LEAD-007	At Lead conversion, if a matching Account already exists, the system shall prompt the user to link to the existing Account rather than create a new one. [ASM]

5.2 Opportunity Management

ID	Requirement
FR-OPP-001	The system shall allow creation and management of Opportunities, each representing a commercial relationship, derived from a Lead. [REQ]
FR-OPP-002	The system shall support multiple Opportunity types: investment, partnership, service, program enrollment. [REQ]
FR-OPP-003	The system shall track Opportunity stage within a configurable pipeline. [REQ]
FR-OPP-004	The system shall allow assignment of probability, value, and expected closing date to an Opportunity. [REQ]
FR-OPP-005	The system shall maintain a history of stage changes for each Opportunity. [REQ]
FR-OPP-006	The system shall support marking an Opportunity as Closed Won or Closed Lost. [REQ]
FR-OPP-007	When an Opportunity is marked Closed Won, the system shall allow an authorized user to export its data for onboarding into the PMS. [REQ]
FR-OPP-008	The system shall support manual export of handoff data in CSV, Excel, and JSON formats. [REQ]
FR-OPP-009	Exported handoff data shall include Account details (organization name, type, industry, contact information), Primary Contact details (name, role, email, phone), and Opportunity details (type, value, closing date, notes, status). [REQ]
FR-OPP-010	Exported data shall conform to a field-level contract agreed with the PMS (see §10.1); the contract itself is an external dependency, not invented by this specification. [REQ, testability contingent on EXT-006]

5.3 Account Management

ID	Requirement
FR-ACCOUNT-001	The system shall store and manage Accounts (organizations). [REQ]
FR-ACCOUNT-002	The system shall categorize Accounts as: startup, investor, company, government, or partner. [REQ]
FR-ACCOUNT-003	The system shall maintain Account profiles including industry, location, and engagement history. [REQ]
FR-ACCOUNT-004	The system shall link Accounts to Opportunities, Contacts, and Activities. [REQ]

ID	Requirement
FR-ACCOUNT-005	The system shall allow exactly one Contact per Account to be designated as the Primary Contact, defaulting to the first Contact added. [ASM]

5.4 Contact Management

ID	Requirement
FR-CONTACT-001	The system shall allow management of individual Contacts associated with Accounts. [REQ]
FR-CONTACT-002	The system shall store Contact name, role, phone, and email. [REQ]
FR-CONTACT-003	The system shall track interactions with each Contact. [REQ]
FR-CONTACT-004	The system shall allow linking a Contact to multiple Opportunities and Activities. [REQ]
FR-CONTACT-005	Each Contact shall belong to exactly one Account in this baseline. [ASM — extensible to many-to-many later if required]

5.5 Activity Management

ID	Requirement
FR-ACTIVITY-001	The system shall record Activities: calls, meetings, emails, and tasks. [REQ]
FR-ACTIVITY-002	The system shall support scheduling and reminders for Activities. [REQ]
FR-ACTIVITY-003	The system shall maintain Activity history for each Lead, Account, Contact, and Opportunity. [REQ]
FR-ACTIVITY-004	The system shall support assigning Activities to users. [REQ]

5.6 Pipeline Management

ID	Requirement
FR-PIPE-001	The system shall provide configurable pipeline stages. [REQ]
FR-PIPE-002	The system shall visualize Opportunities across pipeline stages. [REQ]
FR-PIPE-003	The system shall support filtering pipelines by Opportunity type. [REQ]
FR-PIPE-004	The system shall seed a default pipeline of: New → Contacted → Qualified → Proposal/Negotiation → Closed Won/Lost, fully editable by a System Administrator. [DEF]

5.7 Communication Management

ID	Requirement
FR-COMM-001	The system shall store communication history: emails, messages, call logs. [REQ]
FR-COMM-002	The system shall allow users to log manual communication notes. [REQ]
FR-COMM-003	Integration with external email systems is deferred to a future phase. [REQ — explicitly optional in source]
FR-COMM-004	The system shall provide a unified view of all communications per customer. [REQ]

5.8 Ticket Management

ID	Requirement
FR-TICKET-001	The system shall allow creation and management of support Tickets. [REQ]
FR-TICKET-002	The system shall categorize Tickets as: technical, service request, or general inquiry. [REQ]
FR-TICKET-003	The system shall assign a Priority to each Ticket: Urgent, High, Medium, or Low. [REQ]
FR-TICKET-004	The system shall track Ticket status: open, in progress, resolved, closed. [REQ]
FR-TICKET-005	The system shall assign Tickets to Support Agents; newly created Tickets default to a shared "Unassigned" queue as the responsible party of record until claimed. [REQ, resolved via DEC]
FR-TICKET-006	The system shall use priority-based triage combined with configurable inactivity monitoring; it shall not enforce fixed response/resolution SLAs. [REQ, terminology clarified by DEC]
FR-TICKET-007	The system shall track the last-activity timestamp for each Ticket; any status change, note, or reassignment updates this timestamp. [REQ + ASM]
FR-TICKET-008	The system shall display inactivity duration for each Ticket. [REQ]
FR-TICKET-009	The system shall highlight Tickets with no recent updates. [REQ]
FR-TICKET-010	The system shall provide visual aging indicators based on inactivity. [REQ]
FR-TICKET-011	The system shall categorize Tickets as recent, aging, or inactive, using configurable thresholds (see §6, BR-001 and the threshold matrix). [REQ + DEC]

ID	Requirement
FR-TICKET-012	The system shall send automated reminders to the assignee at the configured reminder threshold. [REQ]
FR-TICKET-013	The system shall notify designated recipients when a Ticket remains inactive beyond its configured escalation threshold. [REQ]
FR-TICKET-014	The system shall automatically escalate Tickets in "Open" or "In Progress" status that exceed their escalation threshold; escalation is represented as a flag independent of Ticket status, not as an additional status value. [REQ + DEC]
FR-TICKET-015	The system shall allow System Administrators to configure escalation thresholds per Priority tier. [REQ]
FR-TICKET-016	Escalation notification recipients shall be configurable independently of RBAC roles (specific users, CRM roles, or, if confirmed necessary, an organizational reporting-line link) — see §13, EXT-002. [REQ, architecture per DEC]
FR-TICKET-017	The system shall move escalated Tickets into a dedicated "Escalated" view. [REQ]
FR-TICKET-018	The system shall visually highlight escalated Tickets on dashboards. [REQ]
FR-TICKET-019	The system shall allow System Administrators to define escalation thresholds, subject to the rule that each tier's reminder threshold is strictly less than its escalation threshold (BR-001). [REQ + DEC]
FR-TICKET-020	The system shall allow System Administrators to configure escalation notification recipients. [REQ]
FR-TICKET-021	Every Ticket shall have a responsible party of record at all times — either a named Support Agent or the shared Unassigned queue. [REQ, resolved via DEC]
FR-TICKET-022	The system shall highlight unassigned Tickets on dashboards. [REQ]
FR-TICKET-023	The system shall track inactive, aging, and escalated Tickets for reporting, retaining escalation history even after the live escalation flag clears. [REQ + ASM]
FR-TICKET-024	The system shall provide dashboards for monitoring ticket queues and support-team performance. [REQ]
FR-TICKET-025	Escalation notifications shall contain only Ticket ID, Priority, category, and inactivity duration, plus a link requiring CRM authentication to view full Ticket content. [ASM, security-relevant]

5.9 Program/Event Management

ID	Requirement
FR-PROGRAM-001	The system shall track program participation: incubation, training, workshops, via a Program Application entity. [REQ + DEC]
FR-PROGRAM-002	The system shall store application data and track enrollment as a status on the Program Application (Applied → Accepted/Enrolled → Completed/Withdrawn). [REQ + DEC]
FR-PROGRAM-003	The system shall track participant progress and engagement, feeding the associated Account's engagement history. [REQ]
FR-PROGRAM-004	The system shall support event scheduling and participant registration. [REQ]
FR-PROGRAM-005	A Program Application may optionally link to an Opportunity when the program has a commercial dimension; non-commercial programs remain unlinked. [DEC — Model C]

5.10 Reporting & Analytics

ID	Requirement
FR-REPORT-001	The system shall generate reports on Leads, Opportunities, and pipeline performance. [REQ]
FR-REPORT-002	The system shall provide dashboards for customer engagement metrics. [REQ]
FR-REPORT-003	The system shall support exporting reports in CSV, Excel, and JSON. [REQ, format standardized by DEF]
FR-REPORT-004	The system shall support filtering and custom report generation. [REQ]
FR-REPORT-005	Specific dashboard metric definitions are deferred to the reporting implementation phase and are not finalized in this baseline. [EXT — see §13]

5.11 RBAC & User Management

ID	Requirement
FR-RBAC-001	The system shall implement role-based access control. [REQ]
FR-RBAC-002	The system shall support the four defined roles: CRM Manager, Business Development Officer, Support Agent, System Administrator. [REQ]
FR-RBAC-003	The system shall implement the visibility model defined in §3.3. [REQ + DEC]

ID	Requirement
FR-RBAC-004	The system shall allow System Administrators to create, deactivate (soft), and manage user accounts; deactivating a user shall prompt reassignment of their owned records. [REQ + DEF]
FR-RBAC-005	The system shall log authentication and authorization-relevant events for audit purposes (see §9). [DEF]

6. Business Rules

ID	Rule
BR-001	For any Ticket Priority tier, the configured reminder threshold must be strictly less than the configured escalation threshold. [DEC]
BR-002	Ownership of a Lead, Account, Opportunity, or Contact may only be transferred by its current Owner, a CRM Manager, or a System Administrator. [ASM, security-relevant]
BR-003	An Opportunity always belongs to exactly one Account. [REQ, implied]
BR-004	A Ticket's inactivity clock resets whenever a status change, note, or reassignment is recorded against it. [ASM]
BR-005	A Ticket's escalation flag clears automatically upon the next recorded activity after escalation; the historical fact that it was escalated is retained for reporting. [ASM]
BR-006	Converting a Lead always creates or links an Account and a Contact; creating an Opportunity is required only when the engagement is commercial in nature. [ASM]
BR-007	A Program Application may exist without ever being linked to an Opportunity; the reverse (an Opportunity requiring a Program Application) is never true. [DEC]
BR-008	Approved threshold values (\$5.8, FR-TICKET-011/019) are configuration data, changeable by a System Administrator without a requirements change or re-approval cycle. [DEC]

7. Workflows / Use Cases

ID	Workflow	Trigger	Steps	Terminal State
WF-001	Lead-to-Opportunity/ Account conversion	Lead qualified	Lead created → assigned → status progressed → converted per BR-006	Account + Contact created; Opportunity created if commercial

ID	Workflow	Trigger	Steps	Terminal State
WF-002	Opportunity pipeline progression	Opportunity created	Stage changes tracked with history; probability/value/close date maintained	Closed Won / Closed Lost
WF-003	CRM → PMS handoff	Opportunity marked Closed Won	Authorized user manually exports Account + Primary Contact + Opportunity data (CSV/Excel/JSON) per the field contract in §10.1	File exported; PMS-side ingestion confirmation is outside CRM scope
WF-004	Ticket lifecycle	Ticket created	Categorized → prioritized → routed to Unassigned queue or a Support Agent → worked through open → in progress → resolved → closed	Closed
WF-005	Ticket inactivity escalation	Inactivity exceeds reminder/escalation threshold	Reminder to assignee → escalation flag set → notification to configured recipients (redacted per FR-TICKET-025) → moved to Escalated view → dashboard highlight → flag clears on next activity	Escalation flag cleared; historical record retained
WF-006	Program/Event enrollment	Application submitted	Program Application created → status progresses (Applied → Accepted/Enrolled → Completed/Withdrawn) → optionally linked to an Opportunity if commercial	Completed or Withdrawn
WF-007	User deactivation and record reassignment	Administrator deactivates a user	User soft-deactivated → system prompts reassignment of that user's owned records → unresolved records remain visible to CRM Manager in the interim	All records reassigned or explicitly left pending, visible to CRM Manager

8. Data Requirements

8.1 Core Entities (domain-level, not a physical schema)

Lead, Account, Contact, Opportunity, Activity, Communication, Ticket, Program, Program Application, User, Role, Escalation Policy, Escalation Recipient, Notification, Audit Log, Export Record, Reference/Configuration Data (categories, sources, pipeline stages, ticket types).

8.2 Reference/Master Data Ownership

System Administrators own and maintain all reference/master data (categories, ticket types, pipeline stage defaults, escalation policies, lead-source taxonomy). **[DEF]**

Proposed seed values (editable, not fixed): - Lead/Account category: startup, investor, company, government, partner **[REQ]** - Lead source: Referral, Website, Event, Partner Referral, Cold Outreach, Walk-in, Social Media, Other **[DEF]** - Ticket category: technical, service request, general inquiry **[REQ]** - Default pipeline stages: New → Contacted → Qualified → Proposal/Negotiation → Closed Won/Lost **[DEF]**

8.3 Data Retention

No retention or deletion policy is defined in the source requirements. This is an external dependency (§13, EXT-005), not invented here. Interim posture: soft-deletion only, indefinite retention pending a defined policy. **[ASM]**

8.4 Data Volume and Scale Assumption

Assumed scale for this baseline: tens to low hundreds of concurrent users; record volumes in the thousands, not millions. Unverified — flagged explicitly rather than treated as confirmed. **[ASM]**

9. Security & Authorization Requirements

ID	Requirement
SEC-001	The system shall implement the role-based visibility model in §3.3. [REQ + DEC]
SEC-002	The system shall enforce BR-002 (ownership-transfer restriction) at the authorization layer, not merely as a UI convention. [ASM, security-relevant]
SEC-003	Passwords shall be hashed and salted; minimum 10 characters; account lockout after 5 failed attempts for 15 minutes; no forced periodic rotation. [DEF]
SEC-004	Authentication shall use short-lived access tokens (approx. 15 minutes) with refresh tokens; the specific mechanism (local credentials vs. integration with an existing identity provider) is an external dependency — see §13, EXT-004. [ASM + EXT]

ID	Requirement
SEC-005	API endpoints shall apply basic rate limiting on authentication routes and restrict CORS to known application origins. [DEF]
SEC-006	The system shall maintain a field-level audit log (who, when, old value, new value) at minimum for Lead, Opportunity, Account, Contact, and Ticket records. [DEF]
SEC-007	Escalation notifications sent outside the authenticated CRM session shall be redacted per FR-TICKET-025. [ASM, security-relevant]
SEC-008	A System Administrator's configuration access shall not implicitly grant visibility into business-record content; export configuration is limited to structure/field-mapping. [DEC]

10. Integration Requirements

10.1 PMS Handoff

The CRM supports a manual, file-based export to the Tenant/Property Management System (PMS) upon an Opportunity reaching Closed Won. **[REQ]**

Export content is limited strictly to: - Account: organization name, type, industry, contact information **[REQ]** - Primary Contact: name, role, email, phone **[REQ]** - Opportunity: type, value, closing date, notes, status **[REQ]**

Export formats: CSV, Excel, JSON. **[REQ]**

No field-level PMS schema is defined in this baseline. The following must be obtained from the PMS system owner before the Export module can be finalized (see §13, EXT-006): field names, data types, mandatory/optional flags, allowed values for enumerated fields, unique-identifier/matching strategy, PMS-side validation rules, exact CSV/Excel/JSON structure, sample files, character encoding, file-naming convention, and schema-versioning approach.

10.2 Other Integrations (Deferred)

- Email system integration for communication logging (FR-COMM-003) — explicitly future-phase per the source document.
- Any API-level (as opposed to file-based) PMS integration — explicitly future-phase per the source document, and not to be assumed without separate approval.
- Single sign-on / external identity provider integration — pending EXT-004.

11. Non-Functional Requirements

No non-functional requirements were present in the original source document beyond RBAC. The items below are explicitly labeled as assumptions or deferred targets, not confirmed requirements, per the taxonomy in this document's legend.

Category	Status
Performance/scale	[ASM] Tens–low hundreds of concurrent users; thousands of records. Unverified.
Availability	[EXT/Deferred] No target defined; deployment-phase decision.
Backup/recovery (RPO/RT0)	[EXT/Deferred] No target defined; deployment-phase decision.
Browser/device support	[DEF] Modern evergreen browsers (current + previous major version), responsive web layout.
Accessibility	[DEF] Baseline practices: semantic HTML, keyboard navigation, adequate color contrast. No formal conformance level (e.g., WCAG tier) specified.
Logging/monitoring (operational)	[Deferred] To be addressed during hardening phase, distinct from the audit logging in §9.

12. Assumptions & Constraints

12.1 Assumptions (full list)

1. Owner = current assignee (not original creator) for Lead/Account/Opportunity/Contact visibility.
2. **[Security]** Ownership transfer restricted to current owner, CRM Manager, or System Administrator (BR-002).
3. Support Agent read-only Account/Contact context extends to any Ticket visible to them (assigned or shared queue), not assigned-only.
4. Password policy: 10+ characters, standard complexity, 5-attempt lockout/15 min, no forced rotation.
5. Users are soft-deactivated, never hard-deleted; deactivation triggers a reassignment prompt.
6. Audit logging covers Lead, Opportunity, Account, Contact, Ticket at field-change granularity initially.
7. Duplicate detection warns (does not block) on exact email/phone match; extended to Lead-to-Account conversion.
8. Pagination: offset-based, 25 records per page by default.
9. Default pipeline stages seeded as New → Contacted → Qualified → Proposal/Negotiation → Closed Won/Lost.
10. Export and report formats standardized on CSV, Excel, and JSON.
11. Ticket time basis: Urgent/High measured in calendar time; Medium/Low measured in a configurable business-hours calendar (actual hours pending EXT-001).
12. New Tickets default to a shared "Unassigned" queue, flagged on dashboards until claimed.
13. Access tokens expire after ~15 minutes with refresh tokens; basic rate limiting on auth endpoints; CORS restricted to known origins.
14. **[Unverified, flagged]** Scale assumption: tens–low hundreds of concurrent users, thousands (not millions) of records.
15. Browser/device support limited to modern evergreen browsers; responsive web; baseline accessibility.
16. System Administrator owns all reference/master data; configures export structure without needing content visibility.

17. Lead source seed list: Referral, Website, Event, Partner Referral, Cold Outreach, Walk-in, Social Media, Other.
18. Email integration and API-level PMS integration remain future-phase.
19. Backup/RPO, availability, and ops-monitoring targets deferred to deployment-architecture phase.
20. Escalation modeled as an orthogonal flag on a Ticket, not a status value; auto-clears on next activity; history retained.
21. Ticket inactivity clock resets on reassignment or any recorded activity (BR-004).
22. Lead conversion always creates/links an Account and Contact; Opportunity creation only when commercial (BR-006).
23. One Account per Contact in this baseline (extensible to many-to-many later).
24. Primary Contact flag added to the Account–Contact relationship; first Contact added defaults to primary.
25. Program Application modeled as one entity with an Applied → Enrolled → Completed/Withdrawn status lifecycle.
26. **[Security]** Escalation notifications contain no PII or ticket body text — ID, priority, category, inactivity duration, and a login-gated link only (FR-TICKET-025).
27. Role-ranking language uses "Role Tier" in project documentation to avoid collision with Ticket Priority.

12.2 Constraints

- The CRM must not implement office allocation, lease management, or billing functionality — these remain the PMS's responsibility. **[REQ]**
- The CRM must not assume or build an API-level PMS integration without a separate, explicit approval beyond this baseline. **[REQ, per project scope instruction]**

13. External Dependencies

ID	Dependency	Who must answer	Blocks
EXT-001	Ethiopian IT Park's actual business-hours calendar (days, hours, holidays)	HR/Operations	Configuration of Medium/Low escalation timing only
EXT-002	Whether "Team Lead," "Manager," "Supervisor" are real distinct organizational positions separate from the four CRM roles, and whether they require CRM logins	Organizational structure owner	Escalation-recipient configuration only
EXT-003	Which specific Ethiopian IT Park programs are commercial (may link to an Opportunity) vs. non-commercial	Program/operations team	Seed data and default UI behavior only
EXT-004	Whether an existing identity provider (SSO) or transactional email/SMS service already exists at Ethiopian IT Park	IT/Infrastructure	Authentication and notification implementation approach only

ID	Dependency	Who must answer	Blocks
EXT-005	Mandatory data retention/deletion policy, if any	Park administration / legal counsel	Retention-policy configuration only
EXT-006	Full PMS Handoff Data Contract (see §10.1)	PMS system owner/vendor	The Export module specifically; no other module
EXT-007	Whether file-attachment support is needed on Tickets/Activities, and where	Business owner / Support Agent supervisors	Scope of the Attachment feature only
EXT-008	Whether manual "escalate now" by a Support Agent is a genuinely separate, wanted capability alongside automatic escalation	Support Agent supervisors	Ticket-escalation feature completeness only
EXT-009	Whether BDOs are organized into teams/branches warranting shared elevated visibility later	Business/HR/org chart	Future visibility-model extension only
EXT-010	Specific dashboard/report metric definitions	CRM Manager	Reporting module implementation only (Phase 10), not architecture

None of the above block the start of system architecture. Each affects a configuration value, an integration detail, or a specific downstream module, not the core domain model or authorization architecture.

14. Out of Scope

ID	Item
OOS-001	Office allocation [REQ]
OOS-002	Lease management [REQ]
OOS-003	Billing [REQ]
OOS-004	Any tenant/property management functionality [REQ]
OOS-005	Real-time/API-based PMS integration, unless separately approved [DEC, per project scope instruction]
OOS-006	External stakeholder (startup/investor) self-service login [ASM, pending EXT-009-adjacent confirmation; assumed no for this baseline]
OOS-007	Email-system integration for communication logging, this phase [REQ — explicitly optional/future in source]

15. Risks

ID	Risk	Why it matters	Mitigation in this baseline
RISK-001	Record-visibility rules are the foundation for every list/read endpoint in the system	Wrong now, expensive to fix later across every module	Policy-layer authorization design isolates visibility logic in one place, not scattered per-query
RISK-002	Ownership-based visibility can be bypassed by unrestricted reassignment	Undermines the entire visibility model silently	BR-002 restricts who may re-assign ownership
RISK-003	Escalation notifications could leak customer data outside RBAC if recipients are non-CRM users	Data governance exposure	FR-TICKET-025 redacts notification content
RISK-004	PMS Handoff Data Contract is undefined	The one true external integration point could fail silently in production if built on guessed fields	No fields invented; export module build gated on EXT-006
RISK-005	Escalation threshold values are unvalidated against real operational patterns	Could over- or under-escalate in practice, eroding trust in the system	Values stored as configuration (not hardcoded), adjustable without redeployment
RISK-006	Scale assumption (§8.4) is unverified	Infrastructure sizing could be wrong in either direction	Conventional relational design chosen to tolerate reasonable variance without redesign
RISK-007	Organizational meaning of escalation-recipient roles (Team Lead/Manager/Supervisor) is unresolved	Notifications could reach nobody, or the wrong people, if guessed incorrectly	Recipient configuration is a simple, explicit list by default (EXT-002), not dependent on an assumed hierarchy

16. Requirements Traceability Matrix

Requirement Group	Source	Status
FR-LEAD-001-007	Lead Management	[REQ] baseline, dedup/conversion behavior via [DEF]/[ASM]
FR-OPP-001-010	Opportunity Management	[REQ] baseline; FR-OPP-010 testability contingent on EXT-006
FR-ACCOUNT-001-005	Account Management	[REQ] baseline; Primary Contact via [ASM]
FR-CONTACT-001-005	Contact Management	[REQ] baseline; cardinality via [ASM]
FR-ACTIVITY-001-004	Activity Management	[REQ] baseline

Requirement Group	Source	Status
FR-PIPE-001-004	Pipeline Management	[REQ] baseline; seed stages via [DEF]
FR-COMM-001-004	Communication Management	[REQ] baseline; FR-COMM-003 deferred
FR-TICKET-001-025	Ticket Management	[REQ] baseline; escalation mechanics, thresholds, and notification content via [DEC]/[ASM]
FR-PROGRAM-001-005	Program/Event Management	[REQ] baseline; Opportunity relationship via [DEC] (Model C)
FR-REPORT-001-005	Reporting & Analytics	[REQ] baseline; specific metrics via [EXT-010]
FR-RBAC-001-005	RBAC & User Management	[REQ] baseline; visibility model via [DEC]
SEC-001-008	Security & Authorization	[DEC]/[DEF]/[ASM] as annotated in §9
INT (§10)	Integration Requirements	[REQ] baseline; PMS field contract via [EXT-006]
BR-001-008	Business Rules	[DEC]/[ASM] as annotated in §6

Coverage confirmation: All 73 functional/RBAC items, 4 integration statements, and 4 source-stated assumptions from the original Ethiopian IT Park CRM requirements document are represented above, either directly or as the basis for a derived/decision item. No source requirement has been dropped or altered in substance.

17. Acceptance / Baseline Statement

This document is proposed as the **requirements baseline for architecture** (Phase 4 of the project plan).

- All source requirements are preserved and traceable (§16).
- All four load-bearing business decisions (record visibility, ticket escalation thresholds, time basis, Program/Opportunity relationship) are approved (§3.3, §5.8, §5.9, §6).
- No unresolved contradiction remains in the ticket lifecycle, RBAC/visibility model, or Lead-Opportunity-Account-Contact conversion flow.
- No PMS schema has been invented; the field-level contract is explicitly deferred to external discovery (§10.1, EXT-006) without blocking the rest of the system.
- All non-functional items are honestly labeled as assumptions or deferred targets rather than presented as confirmed requirements (§11).
- Ten external dependencies remain open (§13); none block the start of architecture.

Baseline statement: Requirements are sufficiently defined for architecture to begin.

18. Appendices

Appendix A — PMS Handoff Data Contract (Pending Discovery)

Checklist to be completed with the PMS system owner before the Export module's field-level design is finalized: field names; data types; mandatory/optional flags; allowed values for enumerated fields; unique-identifier/matching strategy; PMS-side validation rules; CSV structure (column order, delimiter, header, quoting); Excel structure (sheet naming, formatting, multi-record handling); JSON structure (flat vs. nested, single object vs. array); representative sample files; character encoding (UTF-8 assumed, to confirm — relevant given likely Amharic-script names); file-naming convention; schema-versioning approach.

Appendix B — Decision Log Reference

Full decision-by-decision history (record visibility, escalation thresholds, time basis, Program/Opportunity model, and all supporting engineering defaults and assumptions) is maintained in the project's working documents (SRS v0.1–v0.3) and is available on request; this v1.0 baseline reflects their final, approved state.

Appendix C — Change Basis for v1.0

v1.0 consolidates the working drafts (v0.1: initial extraction and gap analysis; v0.2: gap classification and proposed direction decisions; v0.3: final quality review, privilege/state-machine audit, and terminology consistency pass) into a single clean baseline document, formatted for use as architecture input. No new business decisions were introduced in this consolidation beyond what v0.3 had already approved or assumed.